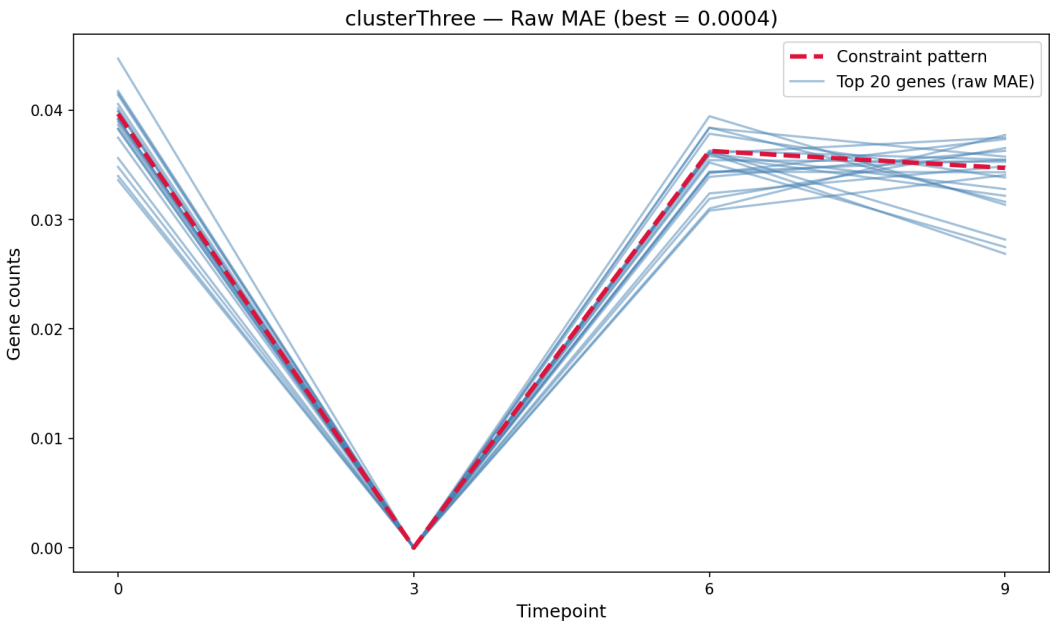
Shape vs. Magnitude

Week of April 23

Binghamton University

April 2026

Where We Left Off (April 16)



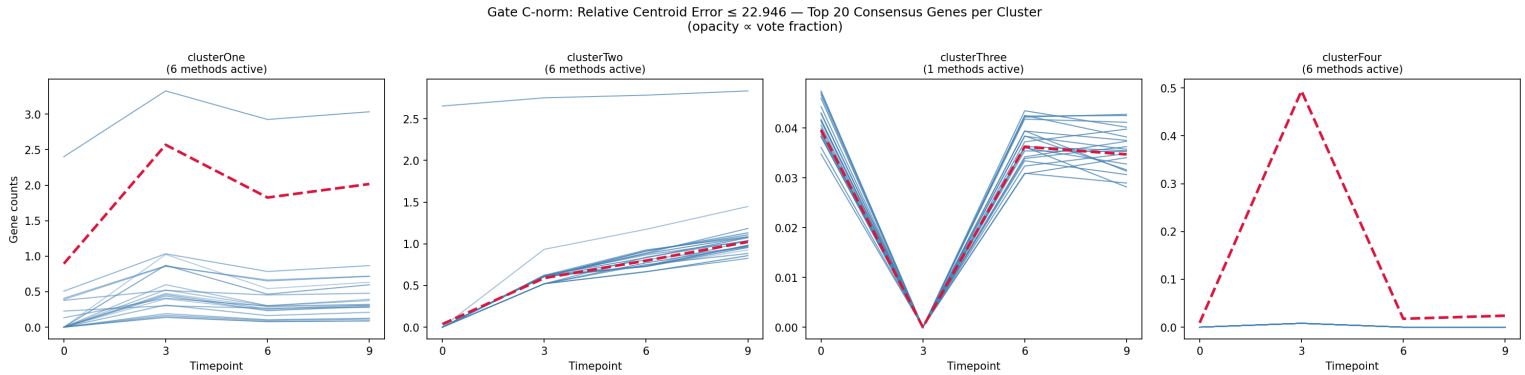
Raw-space MAE gave the best visual fit for clusters Two, Three, and Four.

The six prior methods all operate on *normalized* data — so they find genes with the right *shape* at arbitrary expression levels.

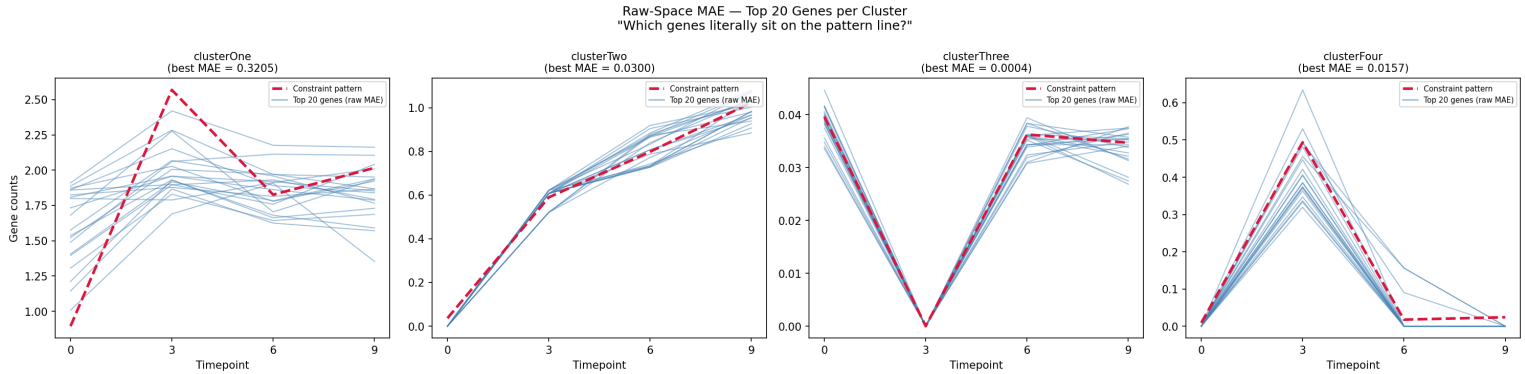
clusterOne was the exception — no genes at the pattern's magnitude also follow its spike shape.

All 4 Clusters — Shape-Based vs. Magnitude-Based

Shape-based gated ensemble



Magnitude-based raw MAE



Question

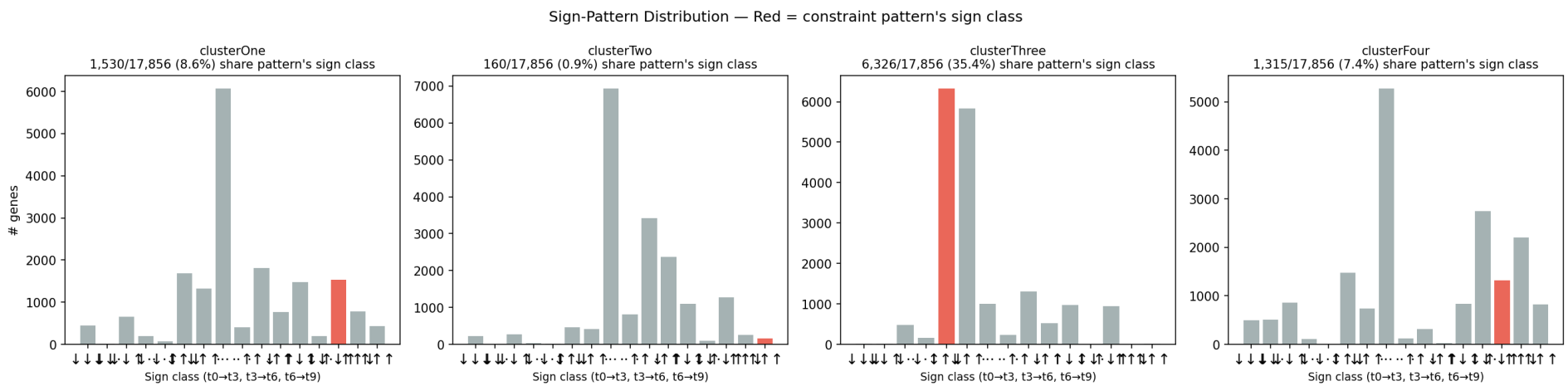
Is biological similarity a matter of

direction of change (shape)

or **absolute expression level** (magnitude)?

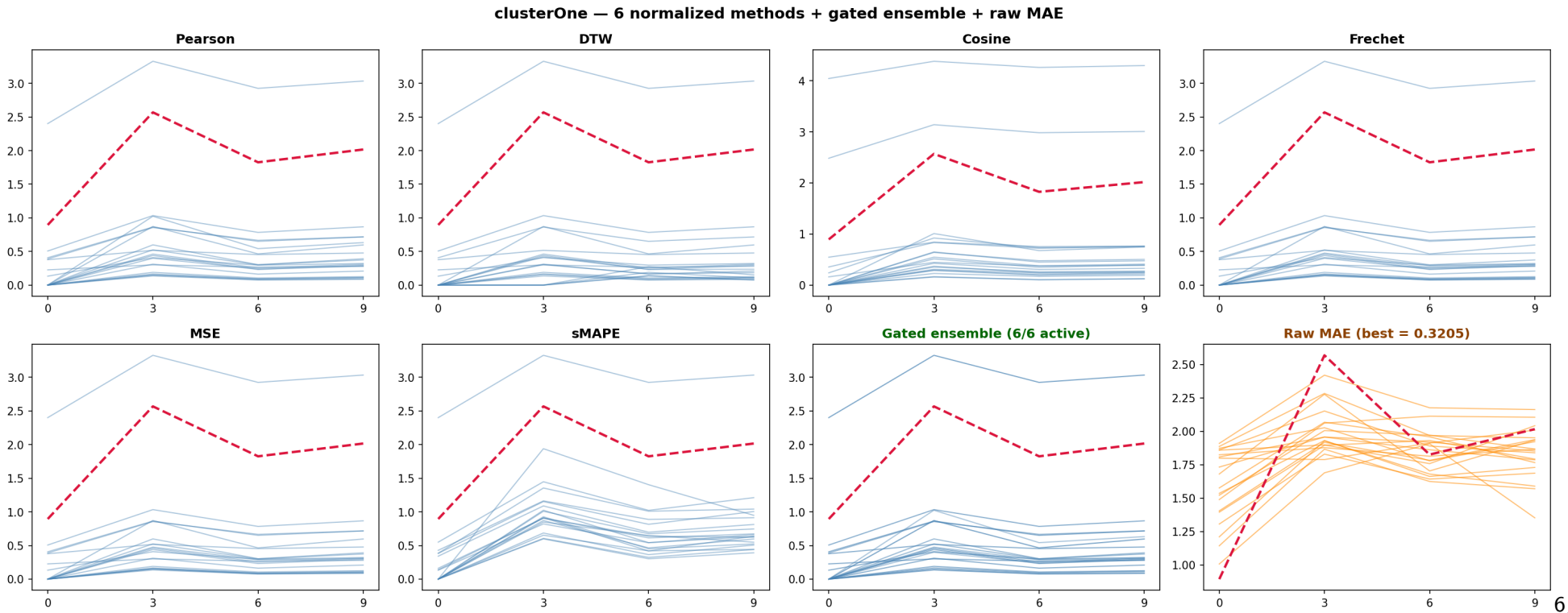
The 4-Timepoint Problem

Each gene's trajectory has 3 transitions ($t_0 \to t_3$, $t_3 \to t_6$, $t_6 \to t_9$). Collapse each to its sign $\Rightarrow$ only $2^3 = 8$ possible "shape classes." All 6 normalized-first methods also erase magnitude, so they're sorting $\sim$17,856 genes into 8 buckets on shape alone. Non-trivial fraction of genes share the constraint's sign class *by accident*.



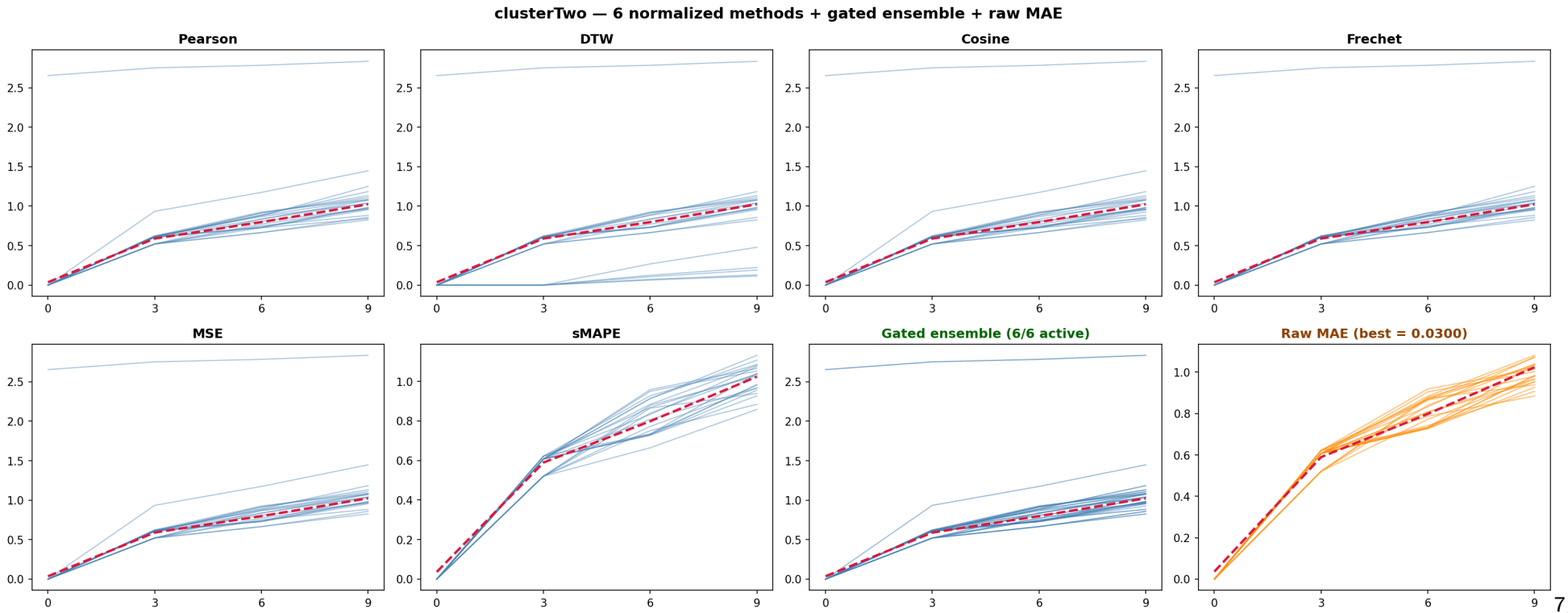
clusterOne — Spike Pattern, Moderate Magnitude

8.6% of genes share the sign class `(+,-,+)`. The 6 normalized methods find spike-like genes mostly *below* the pattern — shape roughly right, magnitude off. Gated ensemble: 6/6 active (no method silenced). Raw MAE: genes sit at the right level but are flat. **Neither path is clean here.**



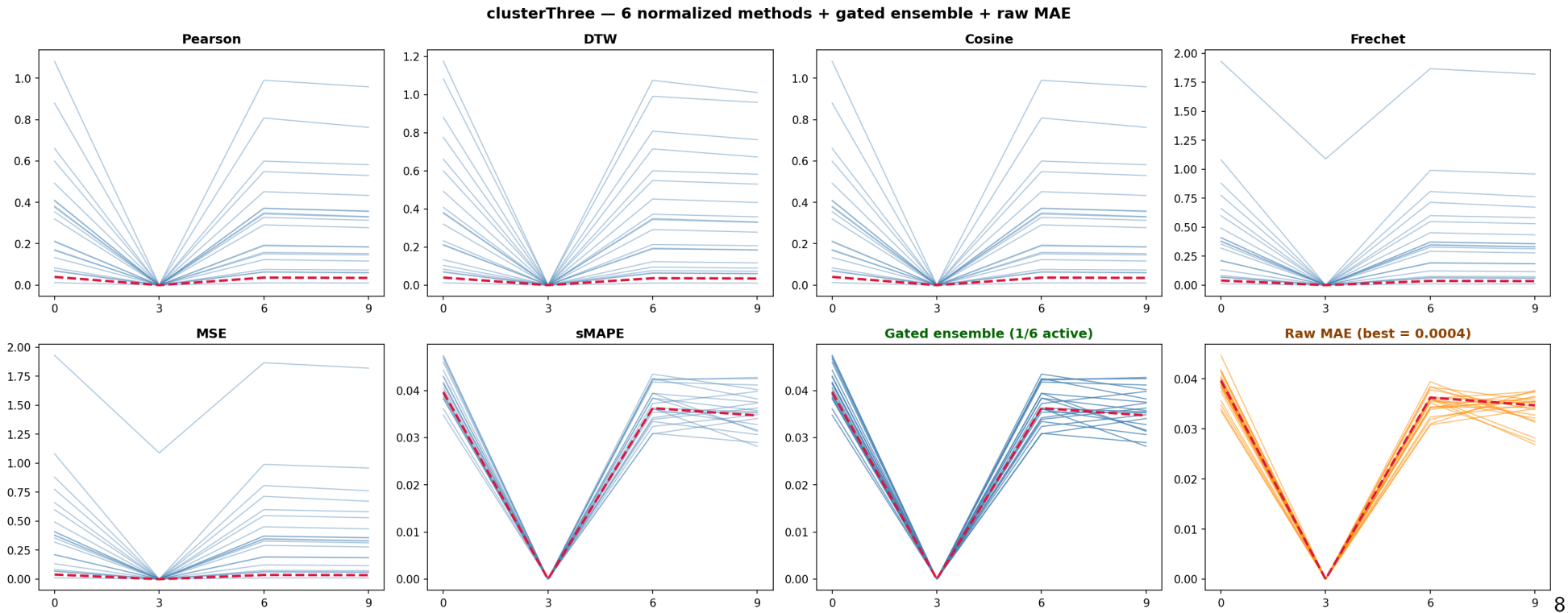
clusterTwo — Monotone Increase

Only 0.9% of genes share (+,+,+) — the rarest sign class, so shape is actually selective here. All 6 methods pick genes that hug both the shape and magnitude. Gated ensemble: 6/6 active, high consensus (RAB14, SYNGR1...). Raw MAE: very tight fit (best ≈ 0.03). **Both paths agree.**



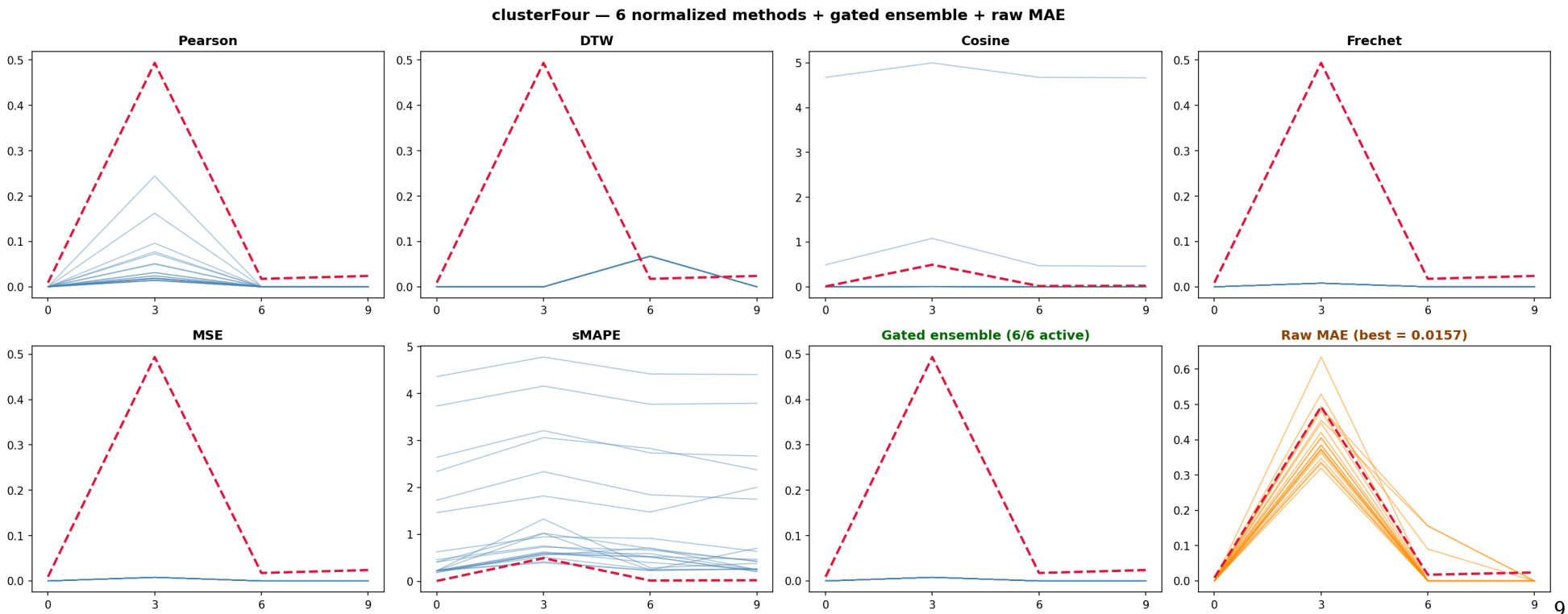
clusterThree — Dip Pattern, Tiny Magnitude

35.4% of genes share `(-,+,-)` — the shape-based filter is weakest here. 5 of 6 normalized methods pick genes 10–40× *above* the pattern (right shape, wrong level). Gated ensemble silences five — only sMAPE survives (1/6 active). Raw MAE: genes sit directly on the pattern. **Magnitude looks closer.**



clusterFour — Sharp Spike, Large Magnitude

7.4% of genes share `(+,-,+)`. 5 of 6 normalized methods pick *flat* genes — normalization erased the spike entirely. Gated ensemble: 6/6 still active — *the gate didn't catch this failure*. Raw MAE captures the spike-shaped genes for the first time. **Magnitude uniquely captures the spike at $t = 3$.**

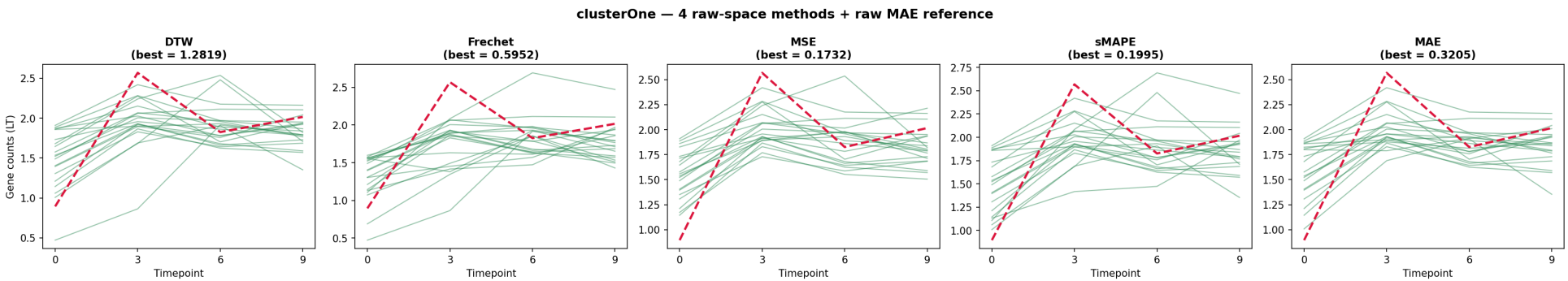


Question for the Group

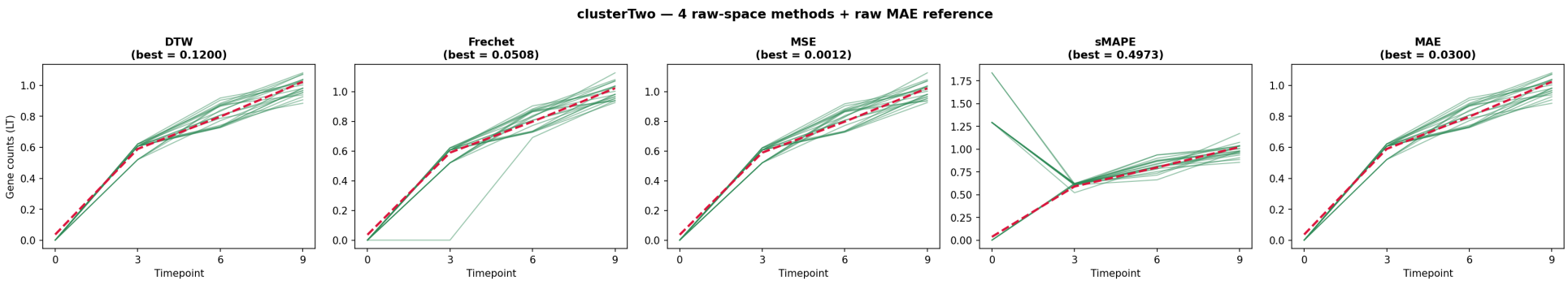
What does "similar gene" mean biologically for these TCR constraint patterns?

Direction of change, or absolute expression level?

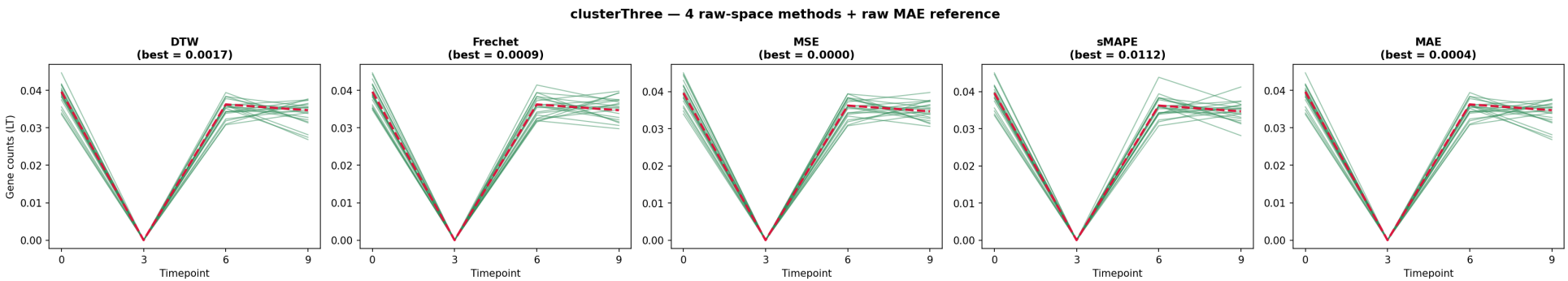
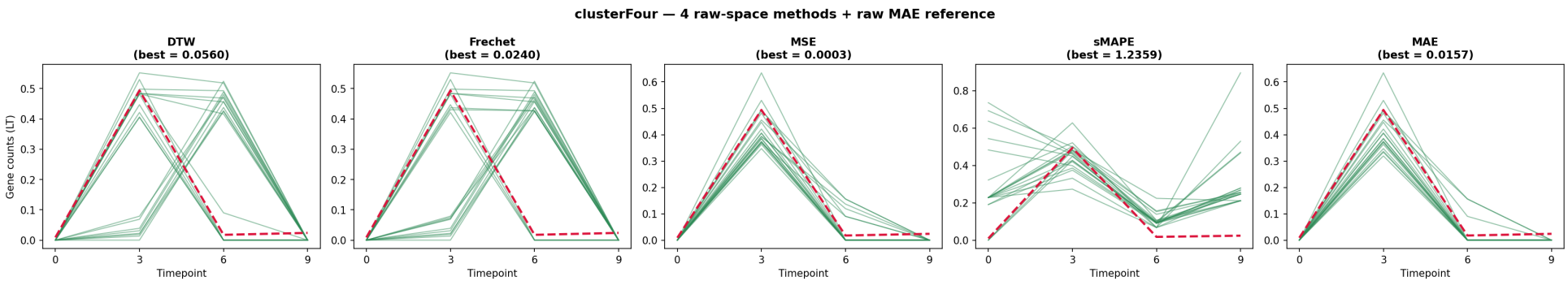
Appendix — clusterOne (raw space)



Appendix — clusterTwo (raw space)



Appendix — clusterThree (raw space)



Appendix — clusterFour (raw space)